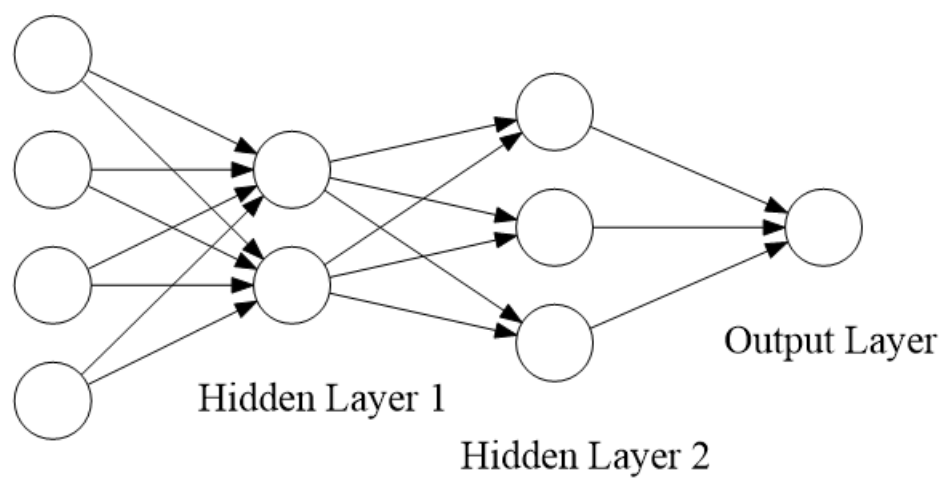


MATH 189 HW 6

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Question 1



a. Input Layer

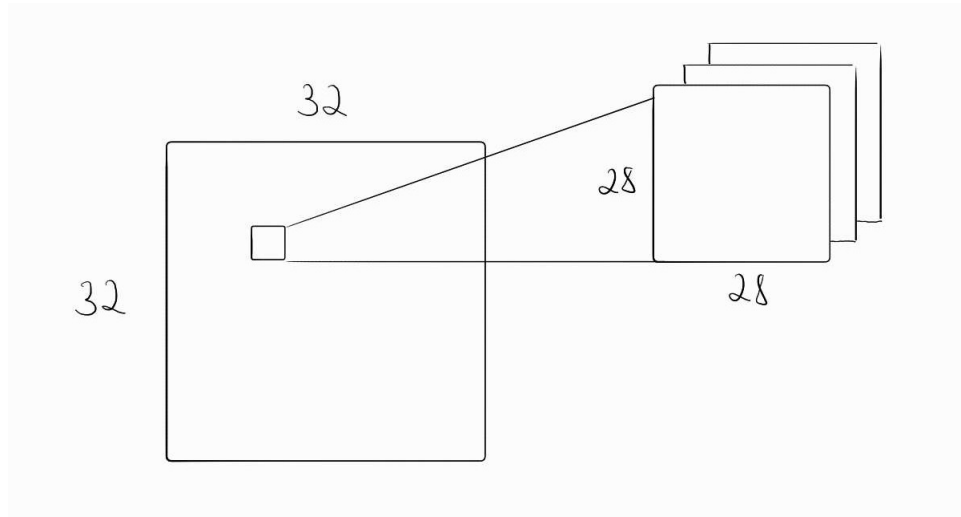
b.

$$f(x) = \beta_0 + \sum_{k=1}^3 \beta_k A_k^{(2)}$$
$$A_k^{(2)} = g \left(w_{k0}^{(2)} + \sum_{j=1}^2 w_{kj}^{(2)} A_j^{(1)} \right)$$
$$A_k^{(1)} = g \left(w_{k0}^{(1)} + \sum_{j=1}^4 w_{kj}^{(1)} x_j \right)$$

c. $f(x) = 1 + 2A_1^{(2)} + 3A_2^{(2)} + 4A_3^{(2)}$

d. $4(2) + 2 + 2(3) + 3 + 3(1) + 1 = 23$ free parameters

Question 2



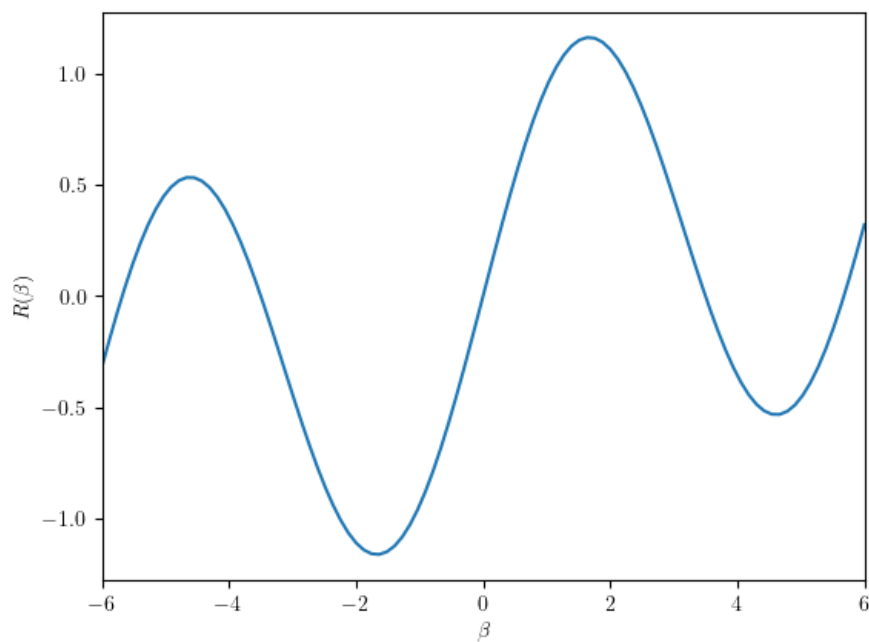
a.

b. There are 3 5×5 filters each with their own bias terms. Therefore, we have $3(5)(5) + 3 = 78$ parameters.

c. The constraints are from the convolution filters being 5×5 , meaning in the input image, only 25 pixels will impact one pixel on the convolved image, that is one constraint.

d. If there were no constraints, there would be $(32 \times 32 + 1) \cdot 3(28 \times 28) = 2,410,800$ parameters.

Question 3

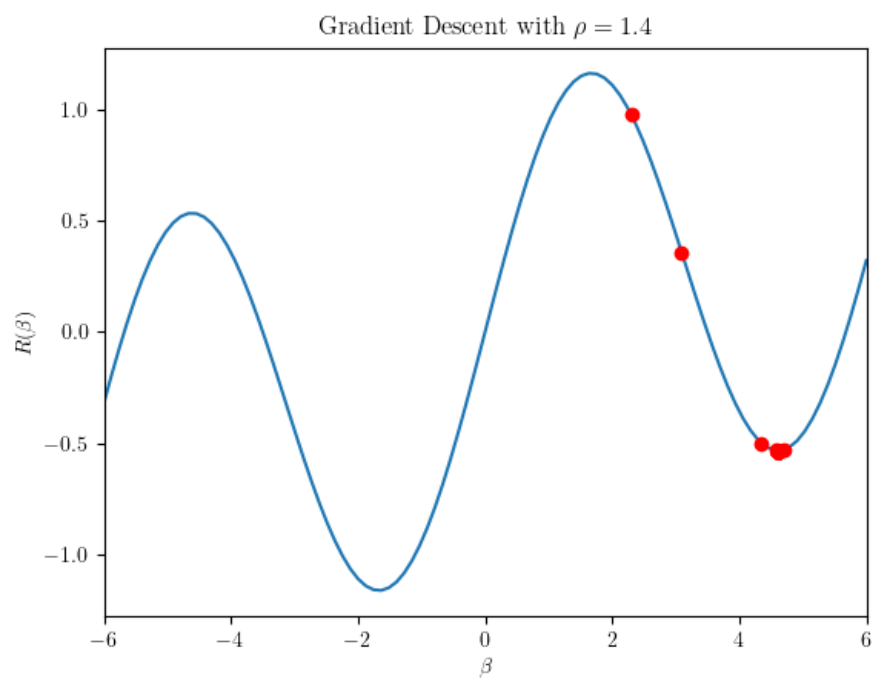
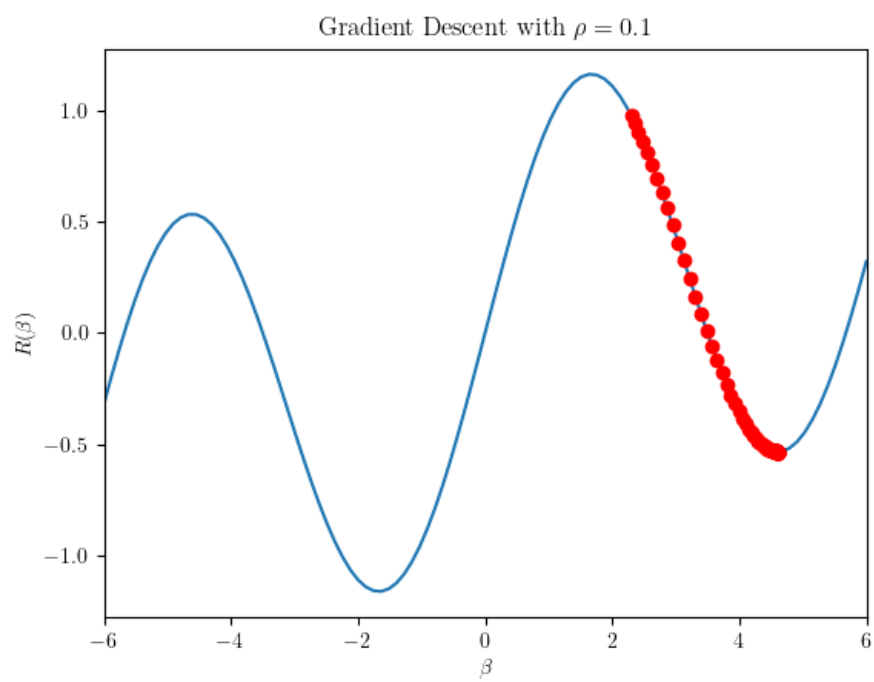


a.

b.

$$\frac{d}{d\beta} R(\beta) = \cos(\beta) + \frac{1}{10}$$

c. I ran gradient descent with $\rho = 0.1$ for 100 steps, $\beta_{100} = 4.612$
With $\rho = 1.4$, $\beta_{100} = 4.612$



Question 4

Skipped